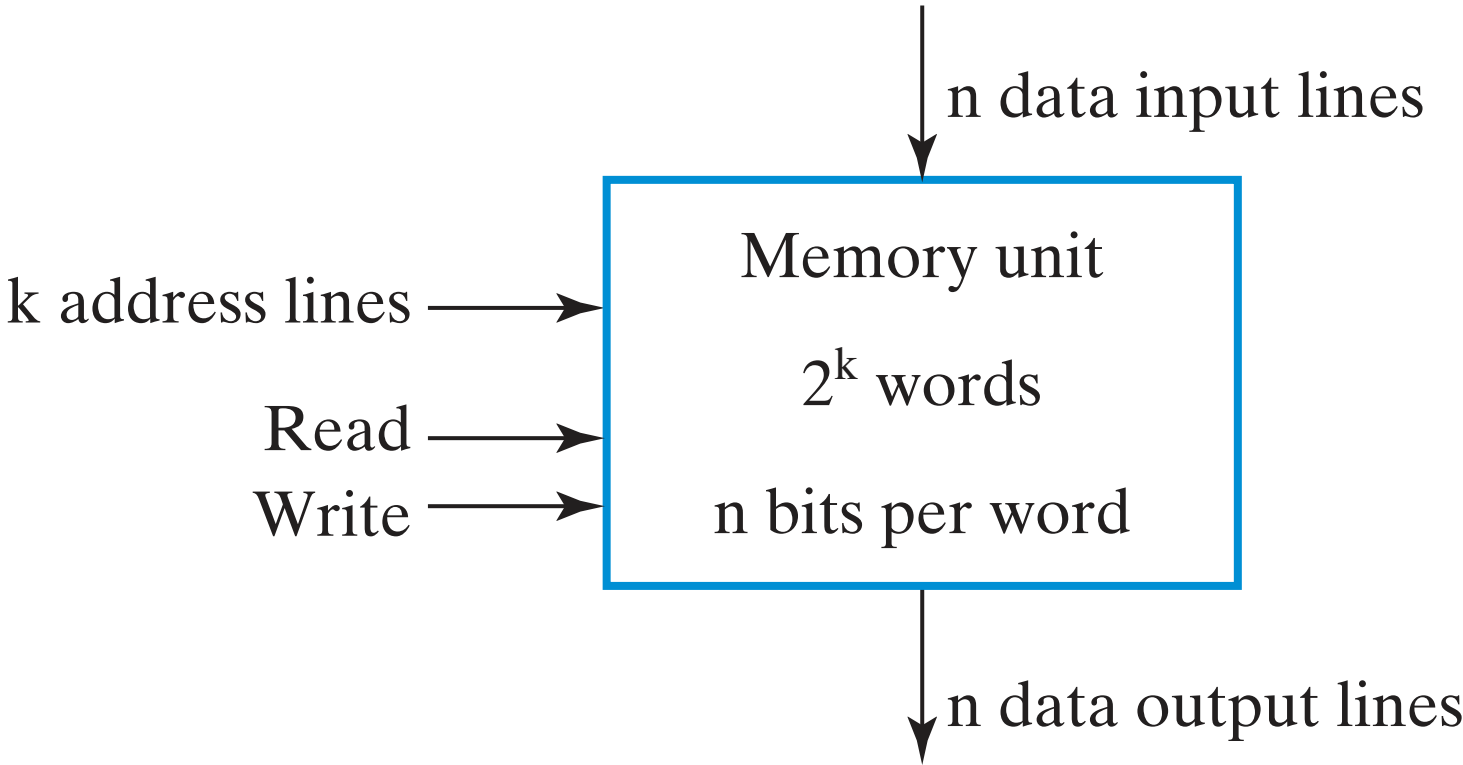
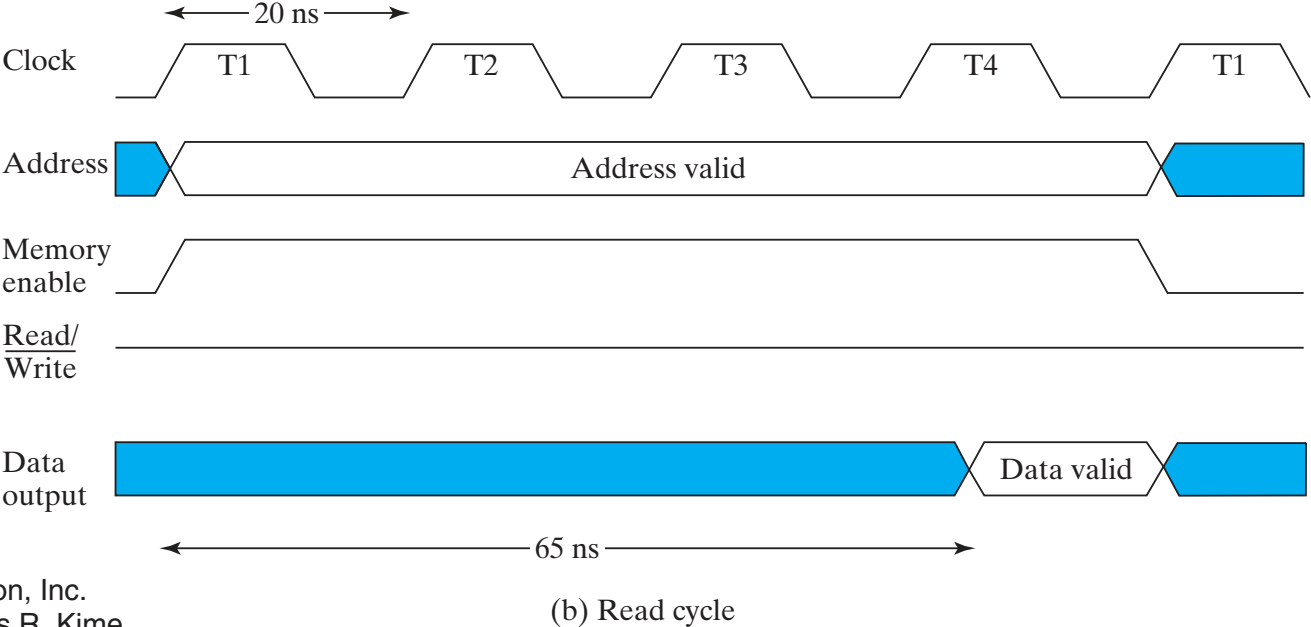
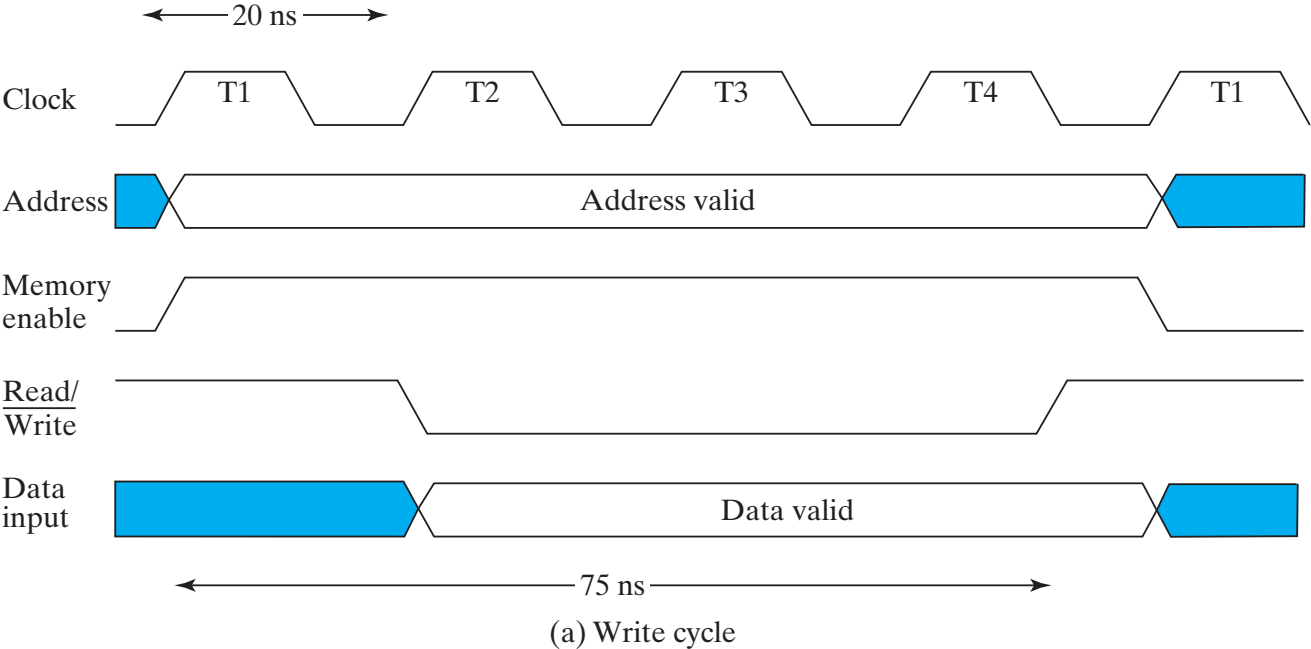


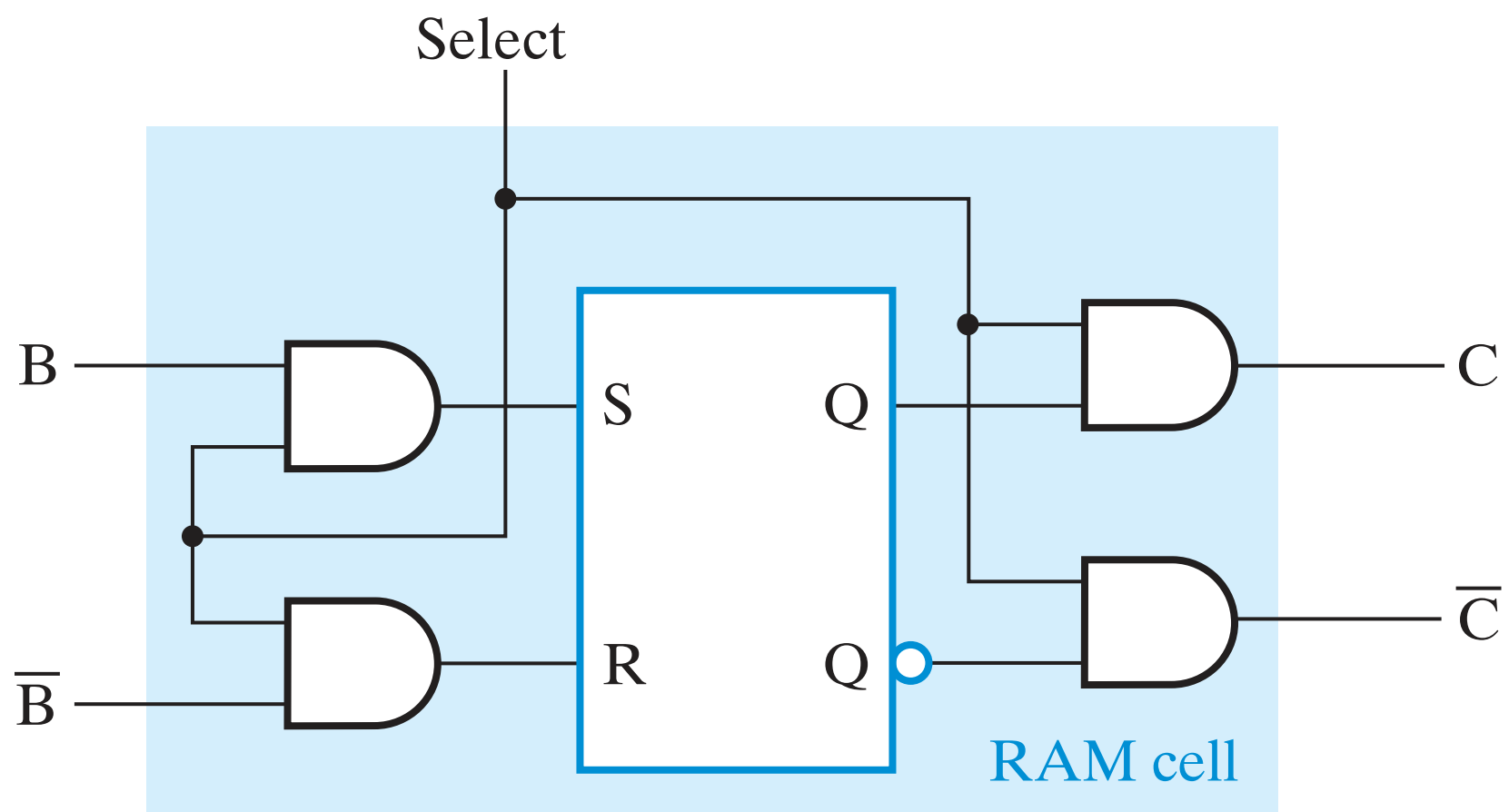


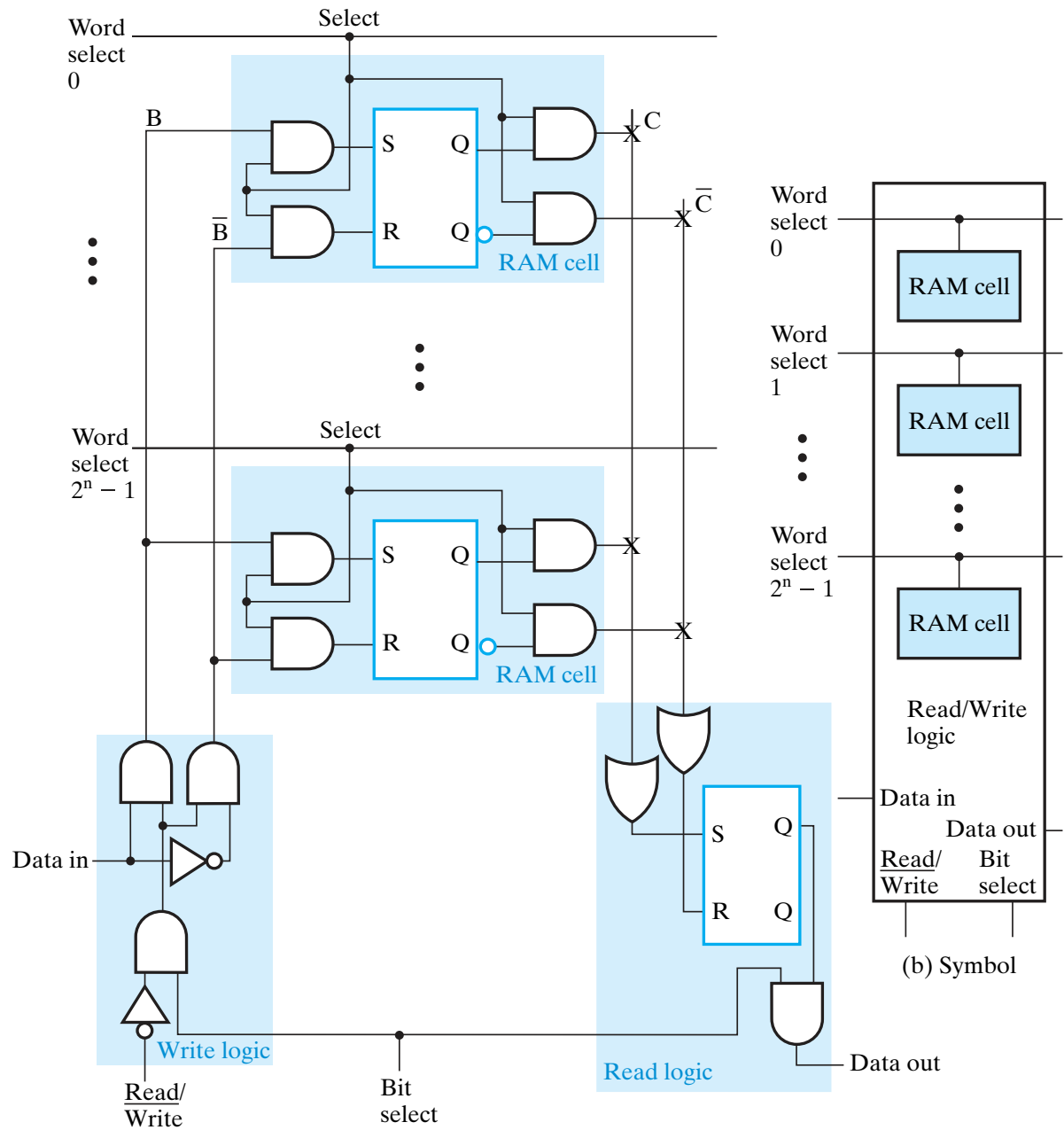
| <u>Memory address</u> |                |                   |
|-----------------------|----------------|-------------------|
| <u>Binary</u>         | <u>Decimal</u> | Memory contents   |
| 0000000000            | 0              | 10110101 01011100 |
| 0000000001            | 1              | 10101011 10001001 |
| 0000000010            | 2              | 00001101 01000110 |
| .                     | .              | .                 |
| .                     | .              | .                 |
| .                     | .              | .                 |
| .                     | .              | .                 |
| .                     | .              | .                 |
| 1111111101            | 1021           | 10011101 00010101 |
| 1111111110            | 1022           | 00001101 00011110 |
| 1111111111            | 1023           | 11011110 00100100 |

**□ TABLE 8-1**  
**Control Inputs to a Memory Chip**

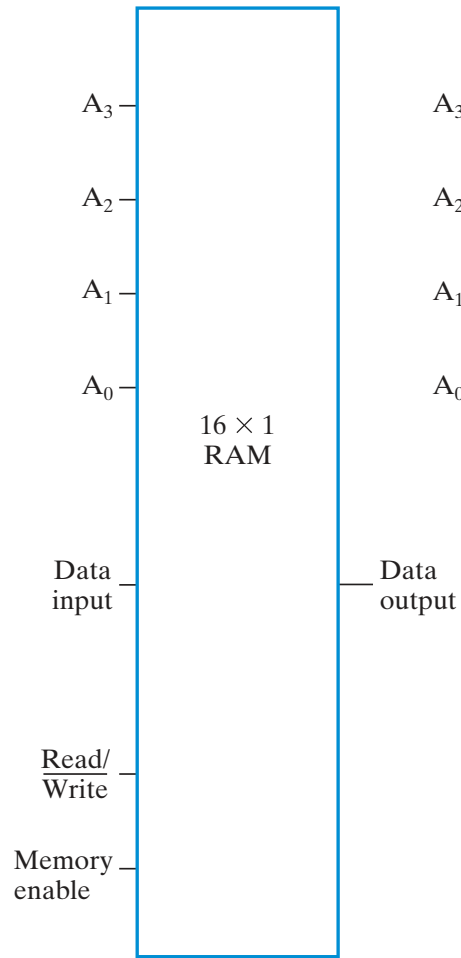
| <b>Chip select</b><br><b>CS</b> | <b>Read/<math>\overline{\text{Write}}</math></b><br><b>R/<math>\overline{\text{W}}</math></b> | <b>Memory operation</b> |
|---------------------------------|-----------------------------------------------------------------------------------------------|-------------------------|
| 0                               | X                                                                                             | None                    |
| 1                               | 0                                                                                             | Write to selected word  |
| 1                               | 1                                                                                             | Read from selected word |



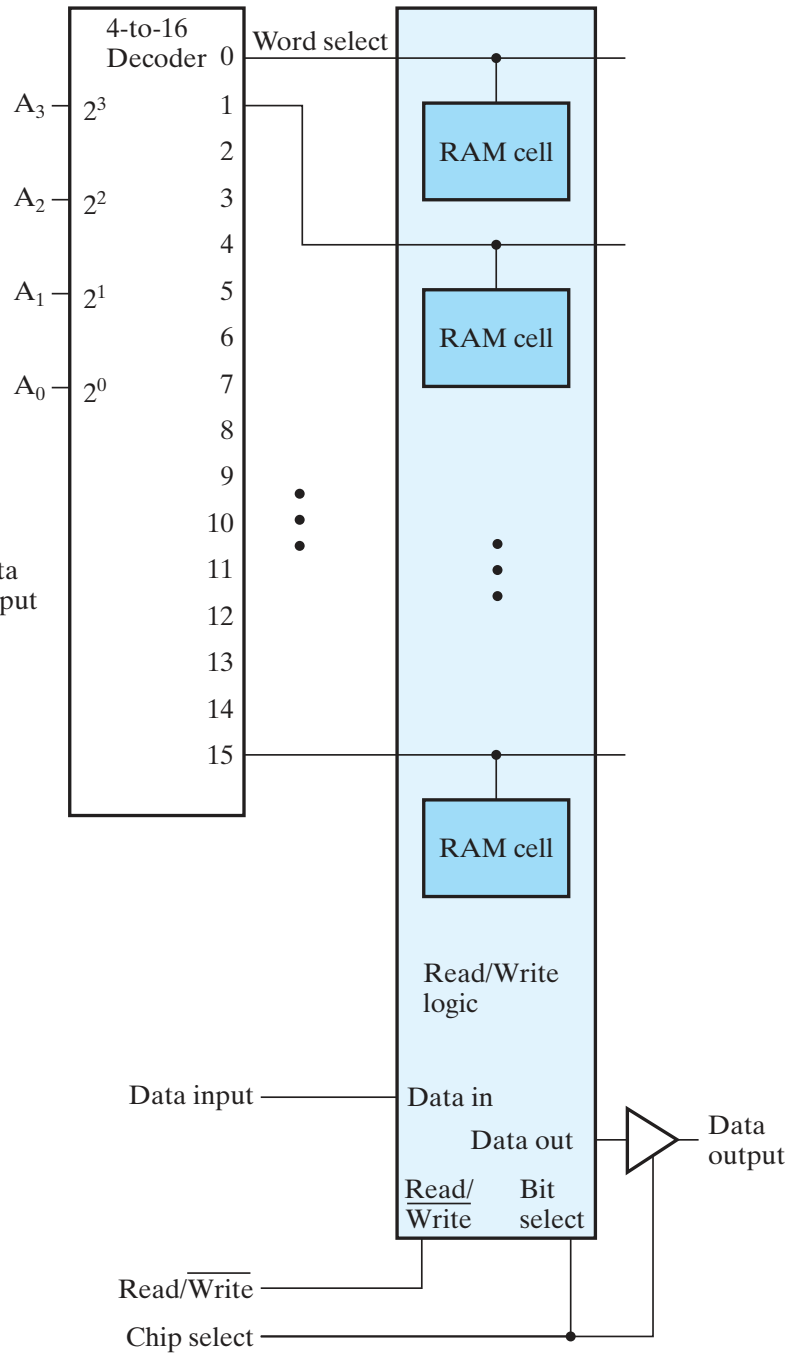




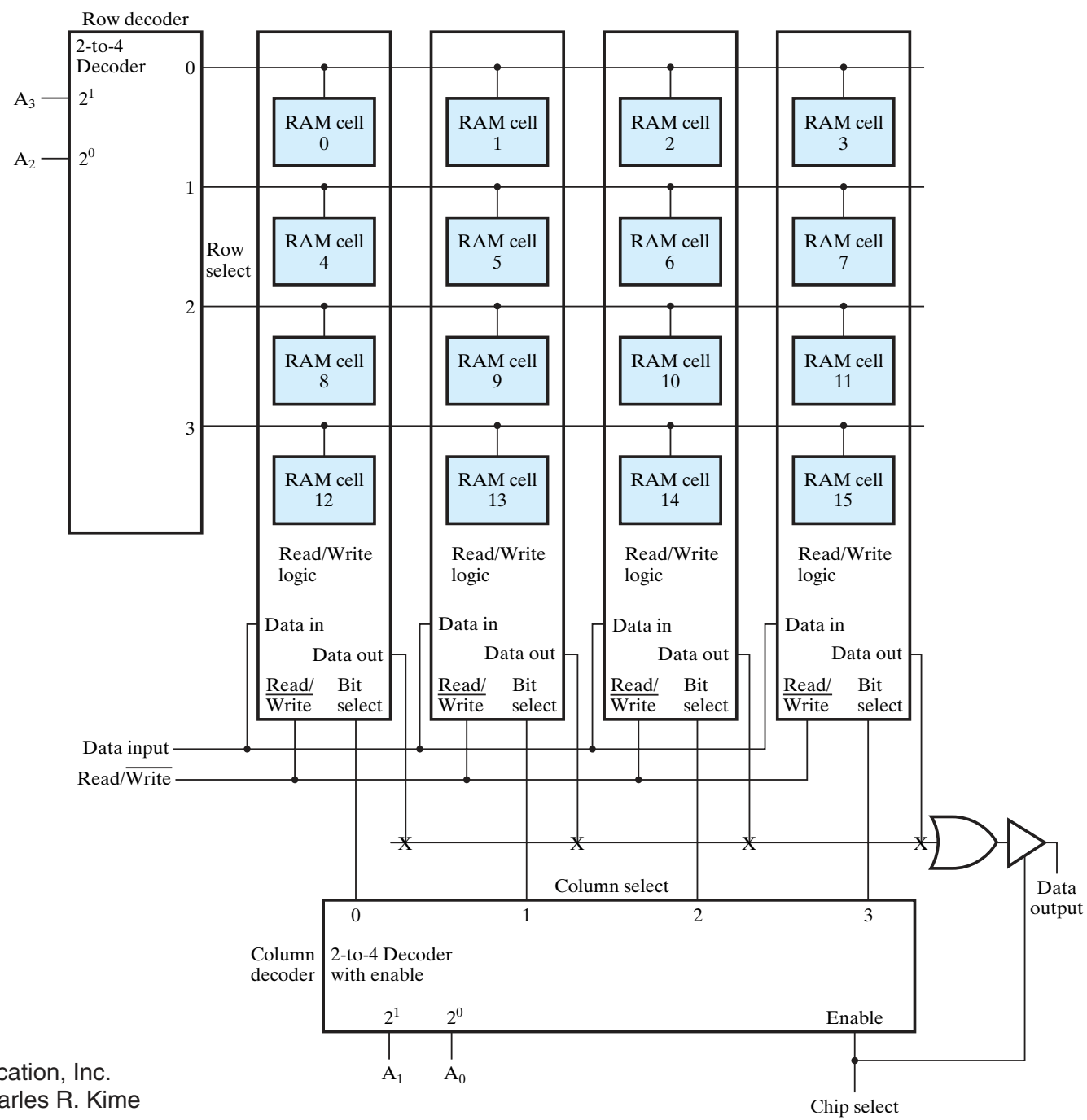
(a) Logic diagram



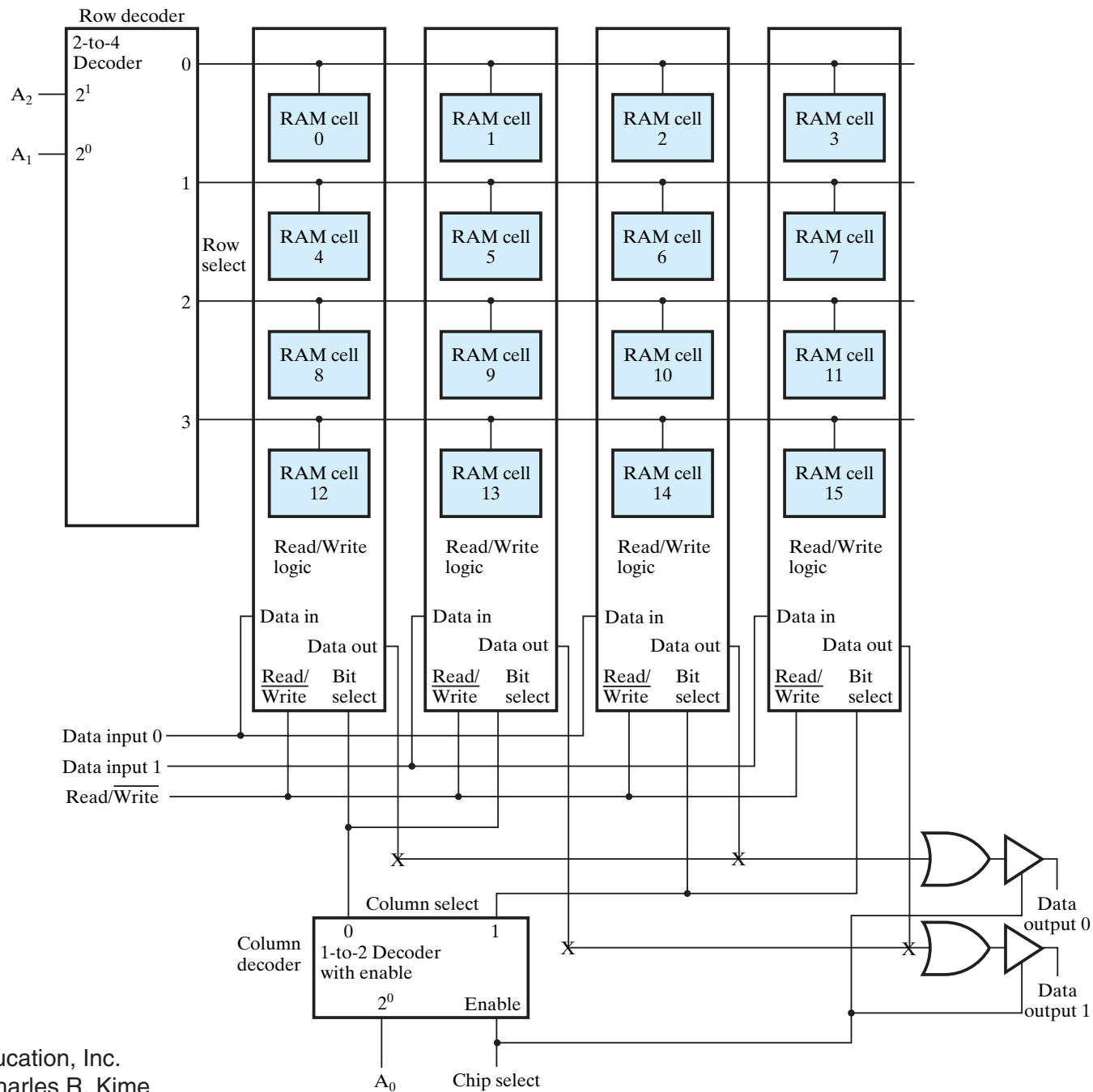
(a) Symbol

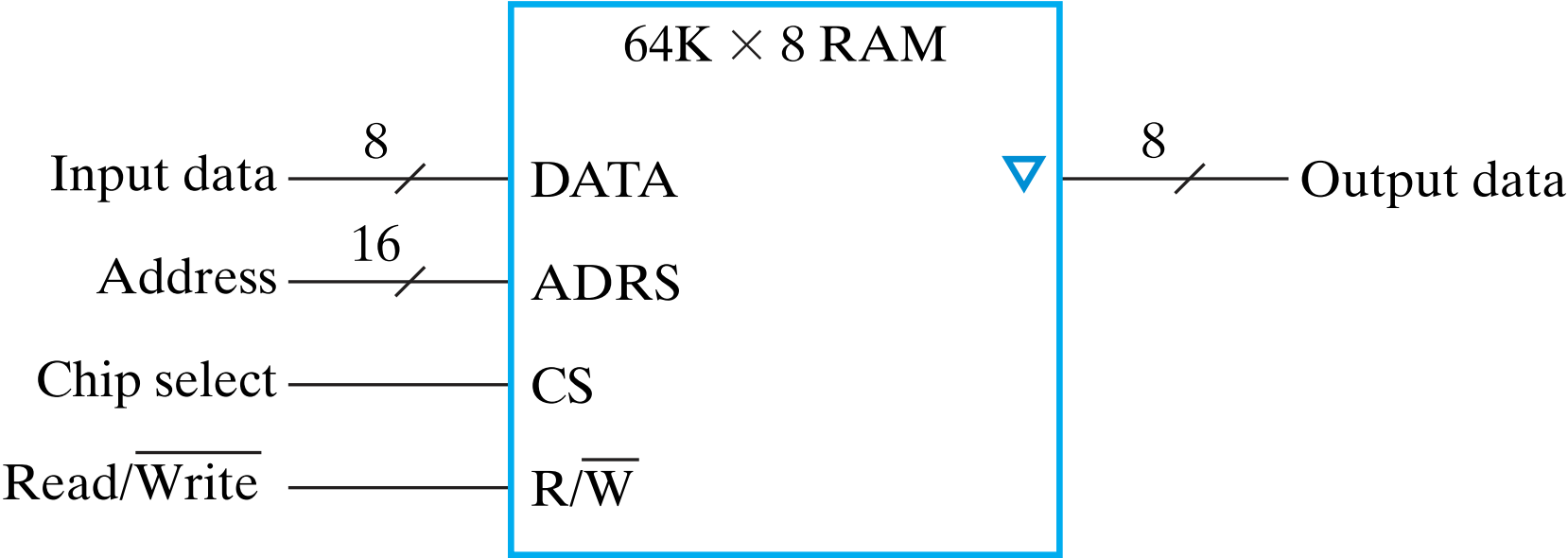


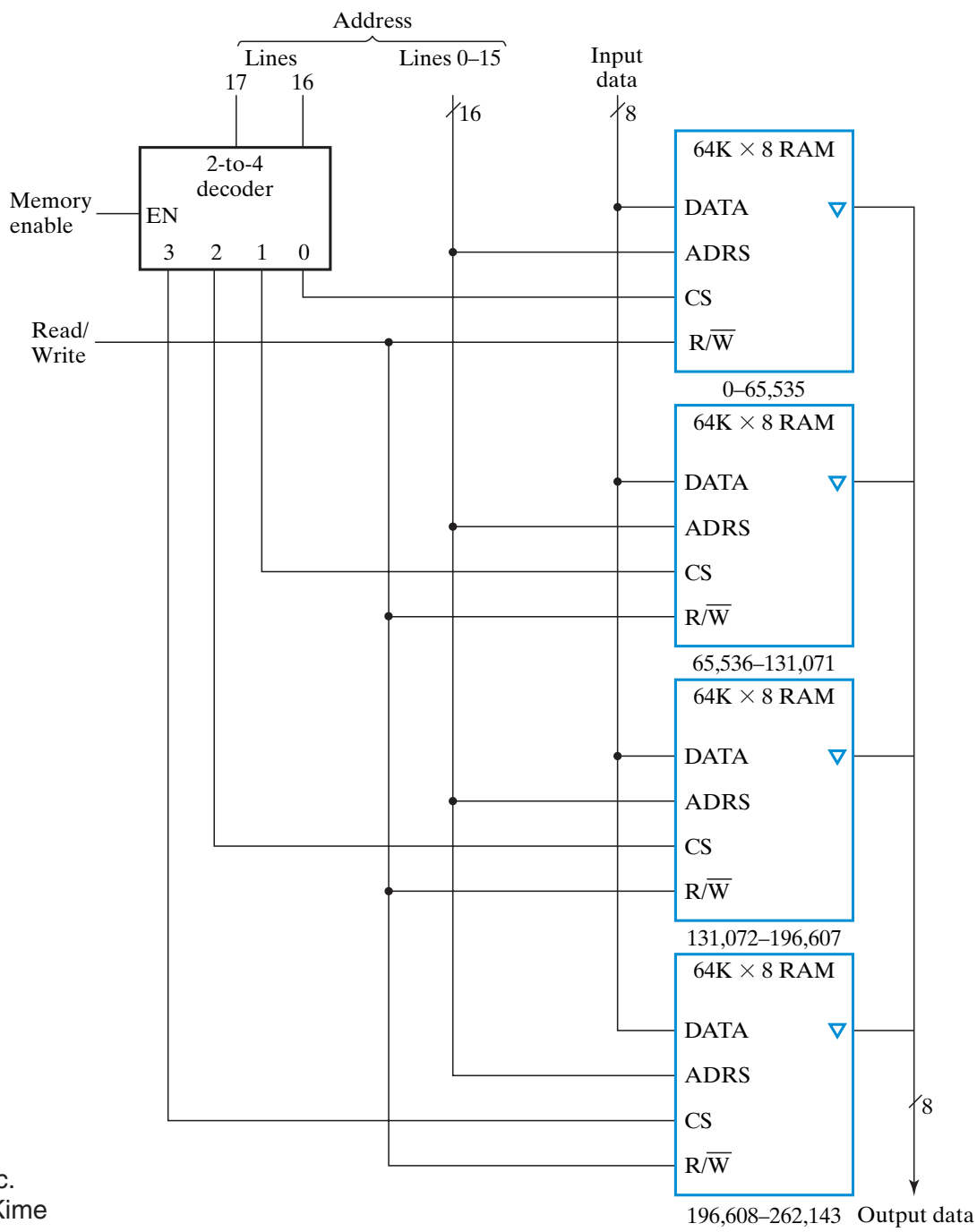
(b) Block diagram

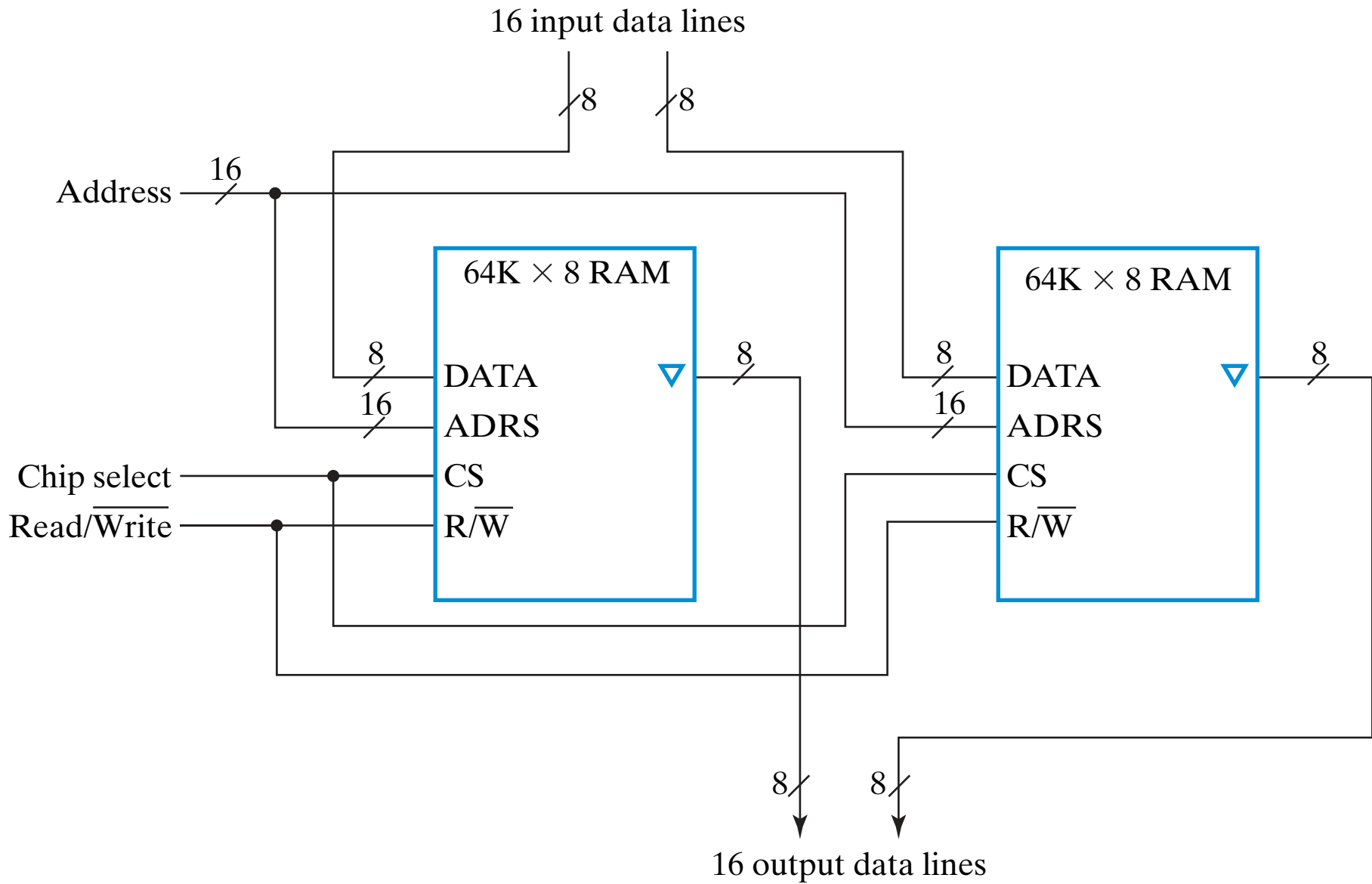


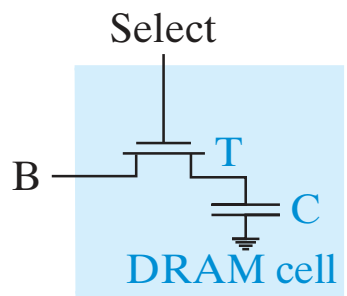






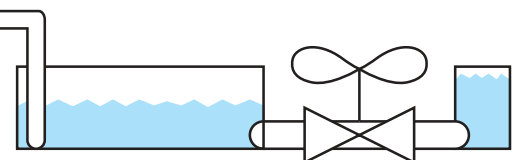




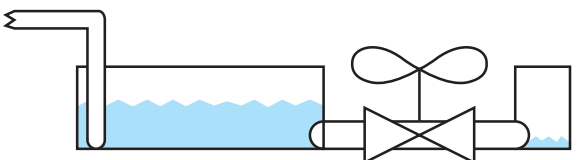


(a)

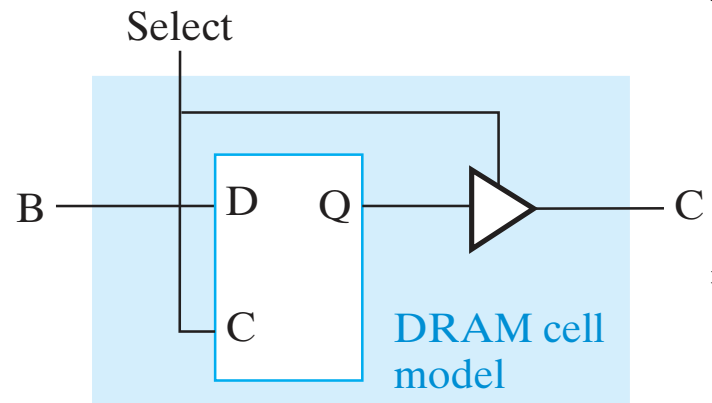
To Pump



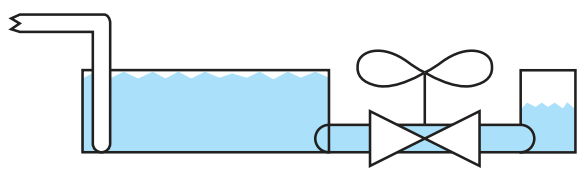
(b)



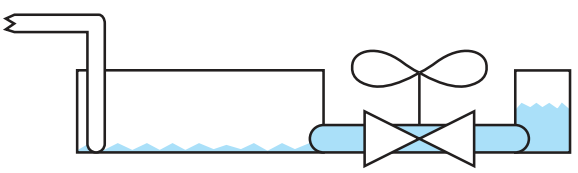
(c)



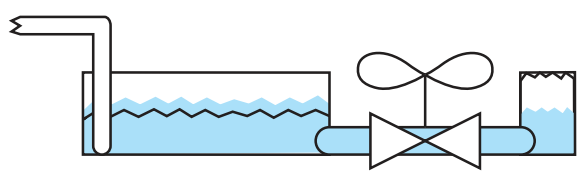
(h)



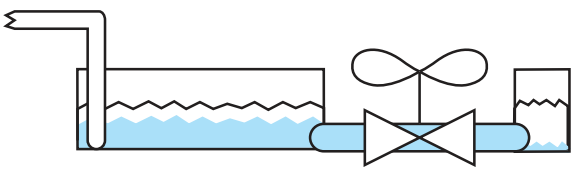
(d)



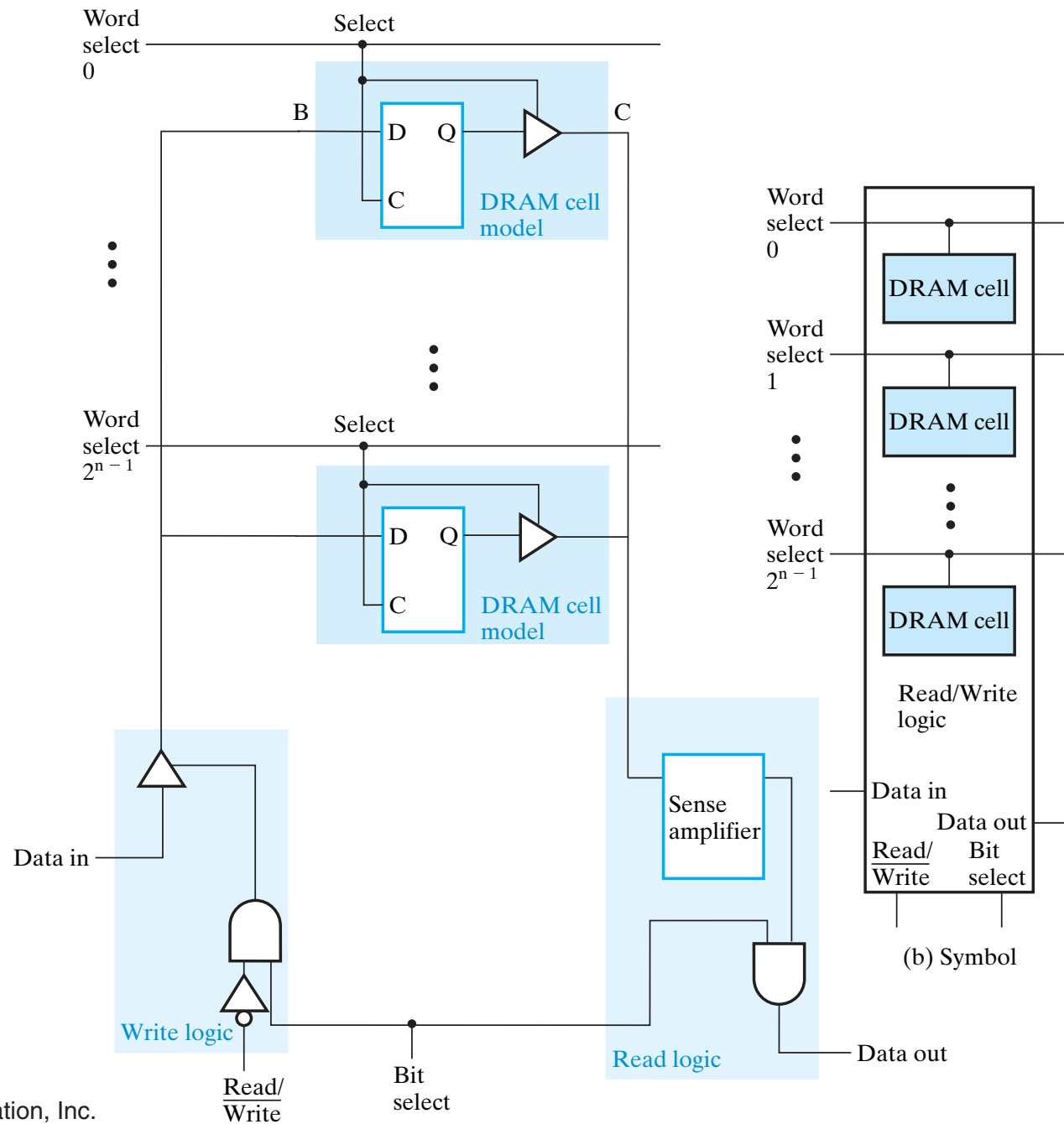
(e)

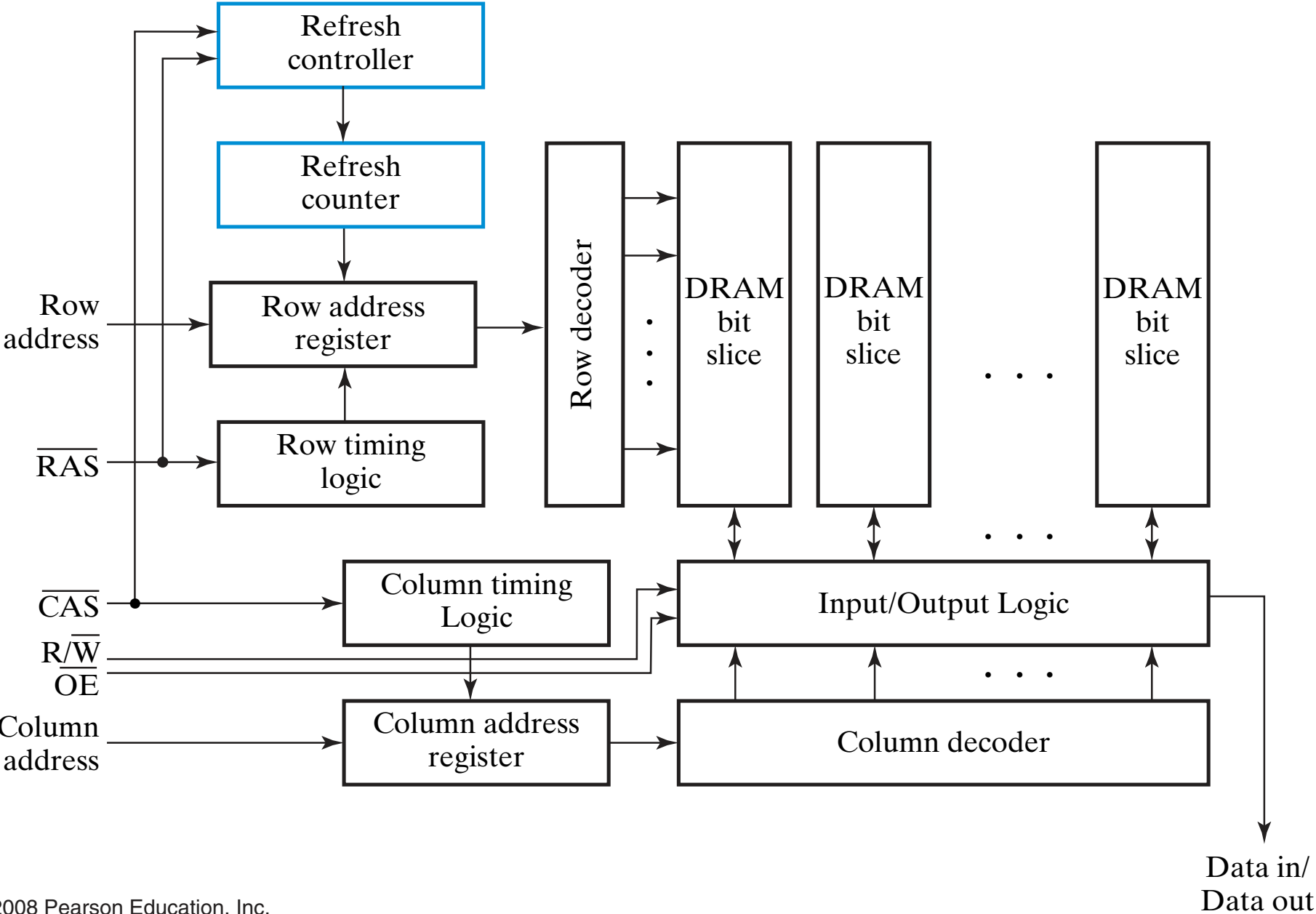


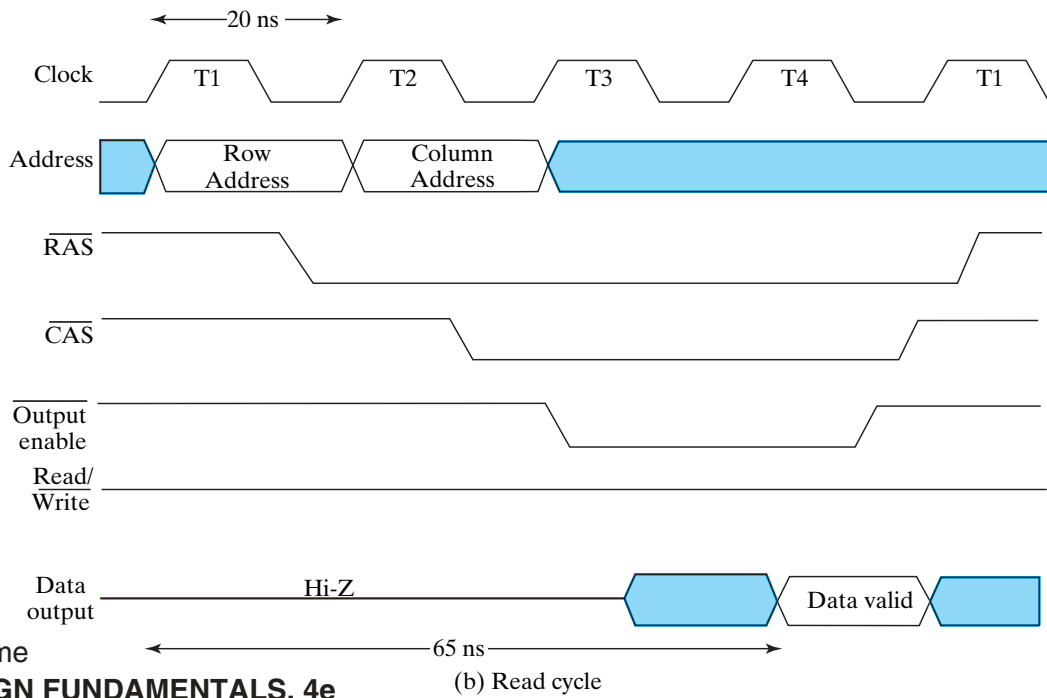
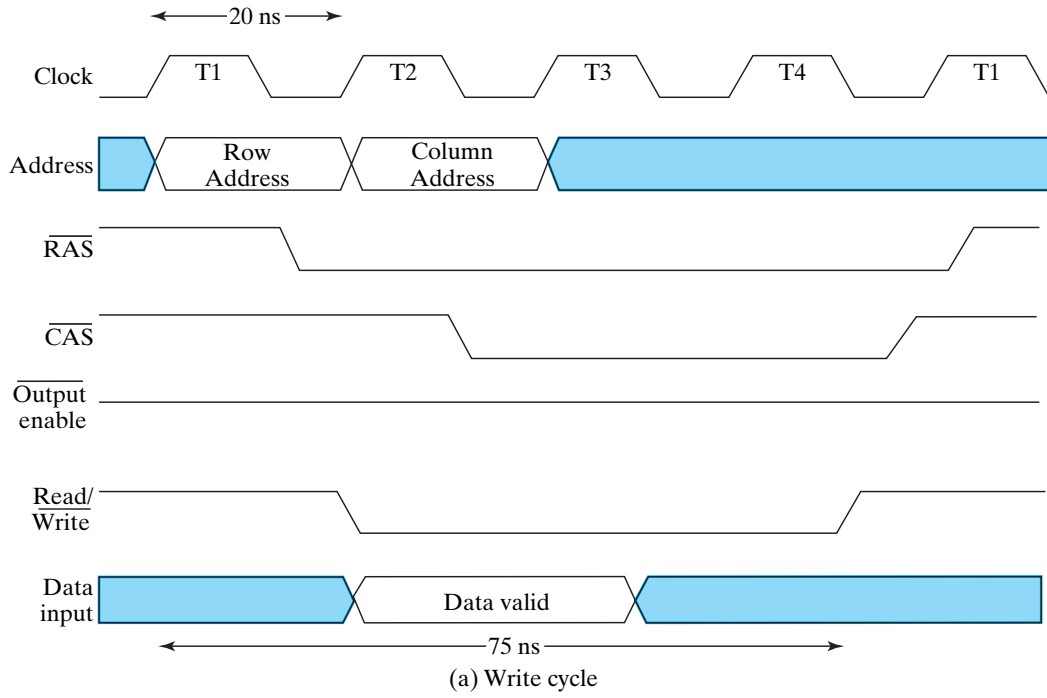
(f)



(g)









**TABLE 8-2**  
**DRAM Types**

| Type                                 | Abbreviation | Description                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fast Page Mode<br>DRAM               | FPM DRAM     | Takes advantage of the fact that, when a row is accessed, all of the row values are available to be read out. By changing the column address, data from different addresses can be read out without reapplying the row address and waiting for the delay associated with reading out the row cells to pass if the row portion of the addresses match. |
| Extended Data Out-<br>put DRAM       | EDO DRAM     | Extends the length of time that the DRAM holds the data values on its output, permitting the CPU to perform other tasks during the access since it knows the data will still be available.                                                                                                                                                            |
| Synchronous DRAM                     | SDRAM        | Operates with a clock rather than being asyn-<br>chronous. This permits a tighter interaction<br>between memory and CPU, since the CPU<br>knows exactly when the data will be available.<br>SDRAM also takes advantage of the row value<br>availability and divides memory into distinct<br>banks, permitting overlapped accesses.                    |
| Double Data Rate<br>Synchronous DRAM | DDR SDRAM    | The same as SDRAM except that data output is<br>provided on both the negative and the positive<br>clock edges.                                                                                                                                                                                                                                        |
| Rambus DRAM                          | RDRAM        | A proprietary technology that provides very high<br>memory access rates using a relatively narrow<br>bus.                                                                                                                                                                                                                                             |
| Error-Correcting<br>Code             | ECC          | May be applied to most of the DRAM types<br>above to correct single bit data errors and often<br>detect double errors.                                                                                                                                                                                                                                |

